

■ 2-MONTH AGGRESSIVE ROADMAP

WEEK 1 — Python for AI (Hands-on only)

Goal: Become comfortable writing clean, async, production-style Python.

Focus

Async/await
File handling (large text files)
Logging
Error handling + retries
Modular project structure
Generators (important for streaming LLM responses)

Daily Tasks

Day 1

Write async script → call 5 public APIs concurrently
Implement retries with exponential backoff manually

Day 2

Build a mini log utility (custom logger with rotating files)

Day 3

Process a 200MB text file (read in chunks, clean, count words)

Day 4

Write a CLI tool: “summarize my file”

Day 5–6 Project:

■ Project #1: Async Web Crawler

Crawl 20 URLs concurrently

Extract text

Save JSON chunks

Add logs + retries

WEEK 2 — LLM Foundations (Pure coding)

Goal: Learn to use LLMs professionally, not toy-level.

Focus

ChatCompletions
Prompts
JSON outputs
Function calling
Streaming responses
Rate limits

Daily Tasks

Day 1: CLI streaming assistant
Day 2: Structured JSON generator
Day 3: Tool-calling agent
Day 4–5 Project:
■ Project #2: Resume Analyzer API

WEEK 3 — Embeddings + VectorDB + Chunking

Goal: Understand retrieval + vector search deeply.

Focus

Chunking
Token splits
Overlap tuning
Embeddings
Vector DBs
Similarity search
Metadata filters

Tasks

Day 1: Chunk PDFs → JSON
Day 2: Embeddings + ChromaDB
Day 3: Metadata filtering
Day 4–5 Project:
■ Project #3: Semantic Search Engine

WEEK 4 — RAG

Goal: Build real RAG systems.

Focus

Query rewriting
Retrieve → rerank → answer
Context injection
Citations
RAG eval

Tasks

Day 1: Basic RAG
Day 2: Query rewriting
Day 3: Reranking
Day 4–6 Project:
■ Project #4: Multi-PDF Chatbot

WEEK 5 — FastAPI Backend

Goal: Convert RAG into API.

Focus

Routers
Dependencies
Pydantic
JWT
WebSockets
Middleware

Tasks

Day 1: CRUD API
Day 2: RAG service class
Day 3: Routes (upload/embed/search/chat)
Day 4–6 Project:
■ Project #5: Full RAG API Service

WEEK 6 — Docker + Cloud

Goal: Deploy like real engineer.

Focus

Docker
Compose
AWS EC2/S3/IAM
Nginx

Tasks

Day 1: Dockerfile
Day 2: Compose
Day 3: Deploy EC2 + Nginx
Day 4–5 Project:
■ Project #6: Cloud RAG System

WEEK 7 — Web UI

Goal: Build a UI.

Focus

Next.js
Streaming UI
Uploads
Auth tokens

Project

■ Project #7: Chat-with-PDFs Web App

WEEK 8 — Final Capstone

■ *AI Knowledge Base System (Production)*

Login
PDF upload
Chunk + embed
Vector DB
Chat UI
Streaming

Citations
History
Logs dashboard
Token usage stats
Cloud deployment
Docker
Monitoring

Outcome After 2 Months

You can:
Build LLM apps
Build RAG systems
Deploy FastAPI
Use vector DBs
Deploy to cloud
Build full-stack AI apps